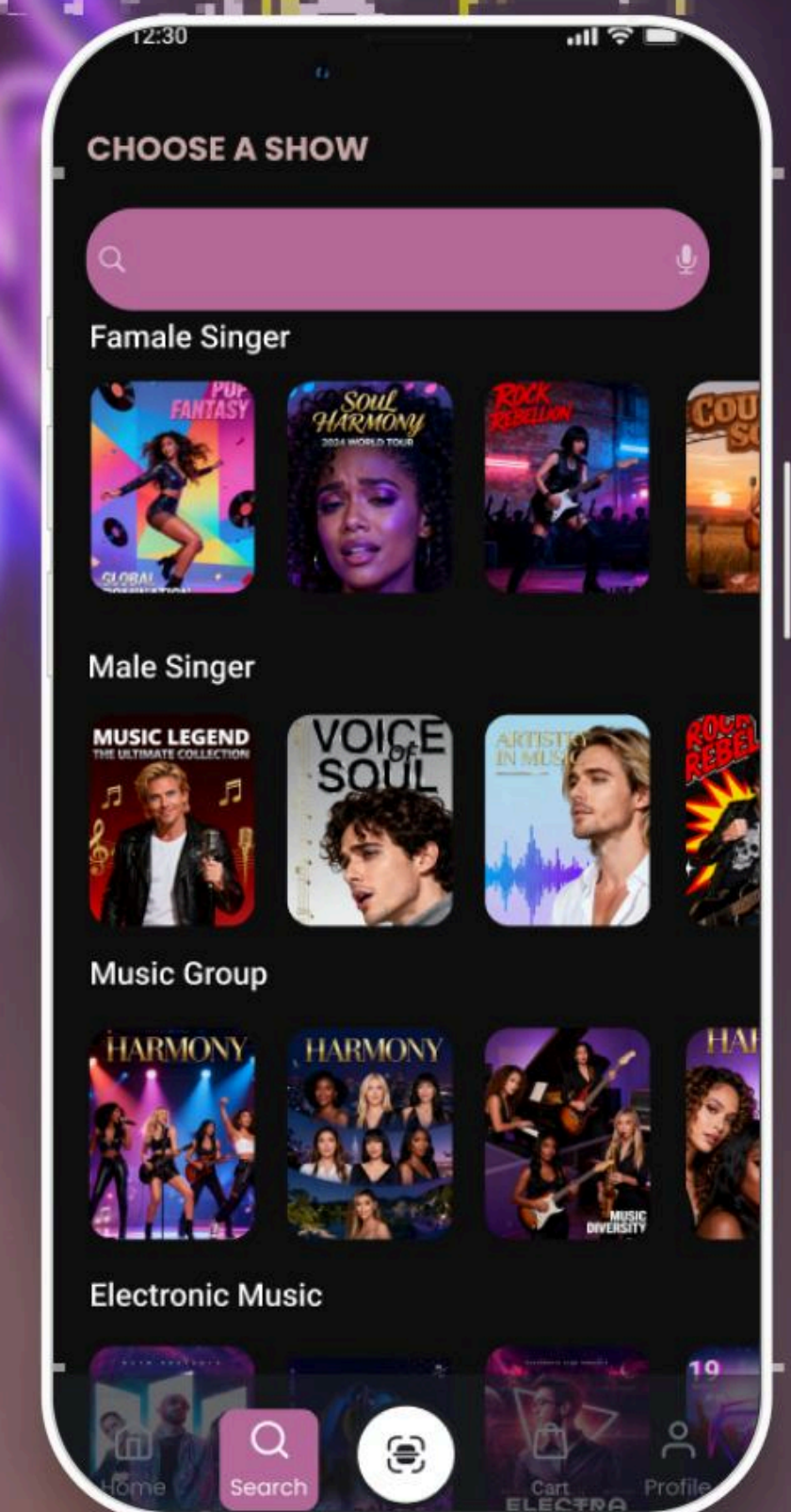


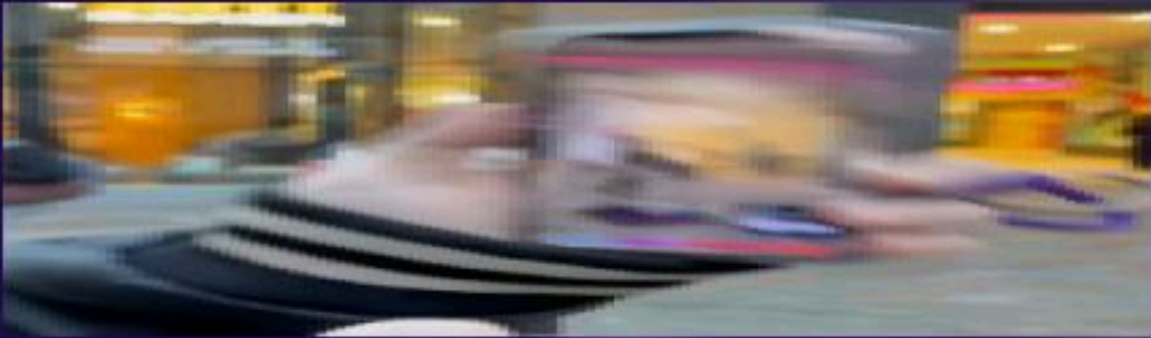
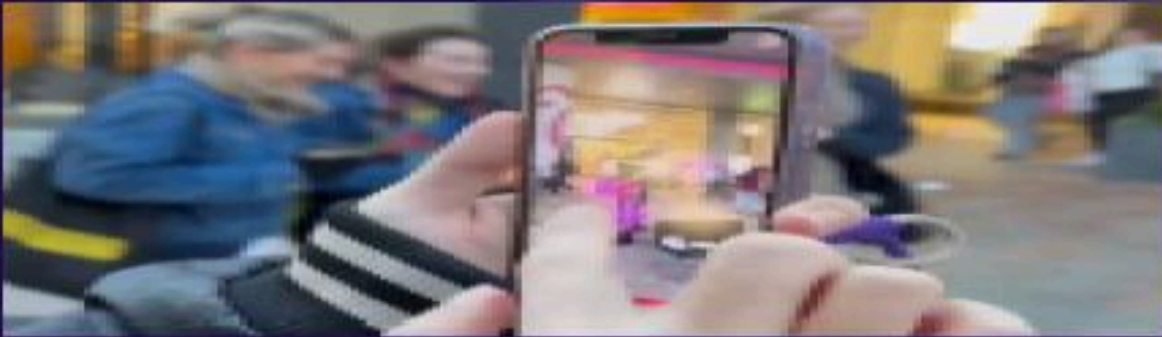
Beat Halo

Let the emotions on stage and in the audience resonate in unison. Let mobile phones cease to be distractions and instead become conduits for connection.

This design aims to enhance interaction between the audience and the stage, increasing participation and immersion. It transforms the phone from a recording tool into a means of engaging in the performance, fostering emotional connection within a narrative setting that doesn't disrupt the stage.



User Interaction Flow

Stage		User goals	Main Interactions	System behavior	Core Experience
Enter the App to prepare for participation			Open the App to access the AR scanning entry point.		
Scan the stage			Align with the stage and scan	Create a shared stage AR	Enter shared interactive space
Rhythm-based interaction lets you express yourself				Shake to generate special effects	Free your hands
Supplementary interactive methods				You can also click and drag to generate special effects.	
Group collaboration	Participation in collectives		When the continuous interaction progress reaches 100%, a grand special effect will be triggered.		Group perception, emotional release
Experience continues	you can record and share		Share to the App		The experience never ends.

Prototype demonstration

```
public class ARItemInteraction_CameraForward : MonoBehaviour
{
    [Header("Refs")]
    public Camera arCamera;
    public ItemManager itemManager;

    [Header("Click Spawn - Random Styles")]
    [Tooltip("点击时会从这里随机挑一个 index 生成。例: 0,4,5,6")]
    public List<int> spawnIndices = new List<int> { 0 };

    [Tooltip("避免连续两次生成同一个样式 (当列表>=2时有效)")]
    public bool avoidRepeat = true;

    [Tooltip("生成距离: 相机前方多少米")]
    public float spawnDistance = 1.2f;

    [Tooltip("允许根据点击屏幕位置在该距离的平面上偏移")]
    public bool useScreenOffset = true;

    [Tooltip("横向/纵向偏移强度 (米)")]
    public float screenOffsetStrength = 0.6f;

    [Tooltip("生成物默认存活时间 (秒)。如果 prefab 自带 ItemAu")]
    public float clickItemLifetime = 5f;

    [Header("UI Block")]
    public bool blockWhenPointerOverUI = true;

    [Header("Debug")]
    public bool debugLog = false;

    public Vector3 LastSpawnCenterWorld { get; private set; }

    int _lastPickedIndex = int.MinValue;

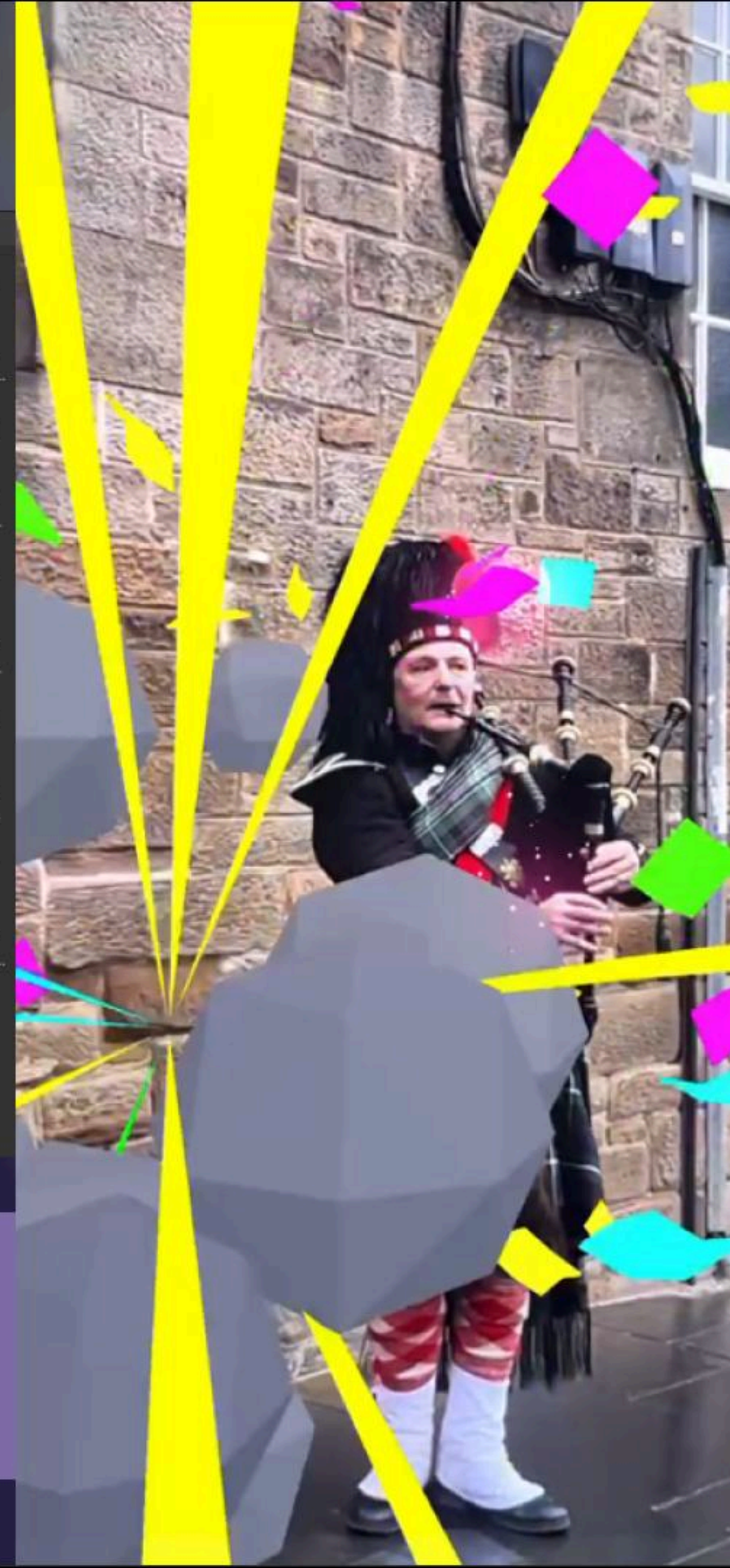
    void Awake()
    {
        if (arCamera == null) arCamera = Camera.main;
        if (itemManager == null) itemManager = FindObjectOfType<ItemManager>();
    }
}
```

Assets > Scripts

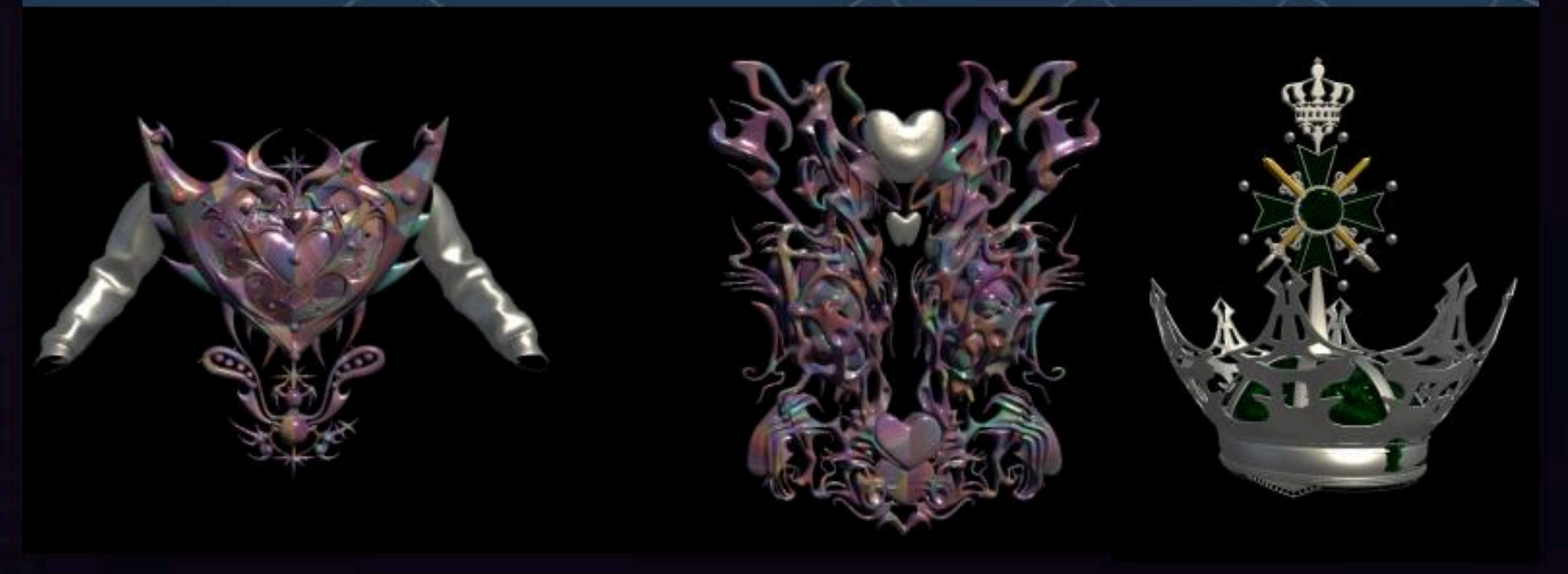
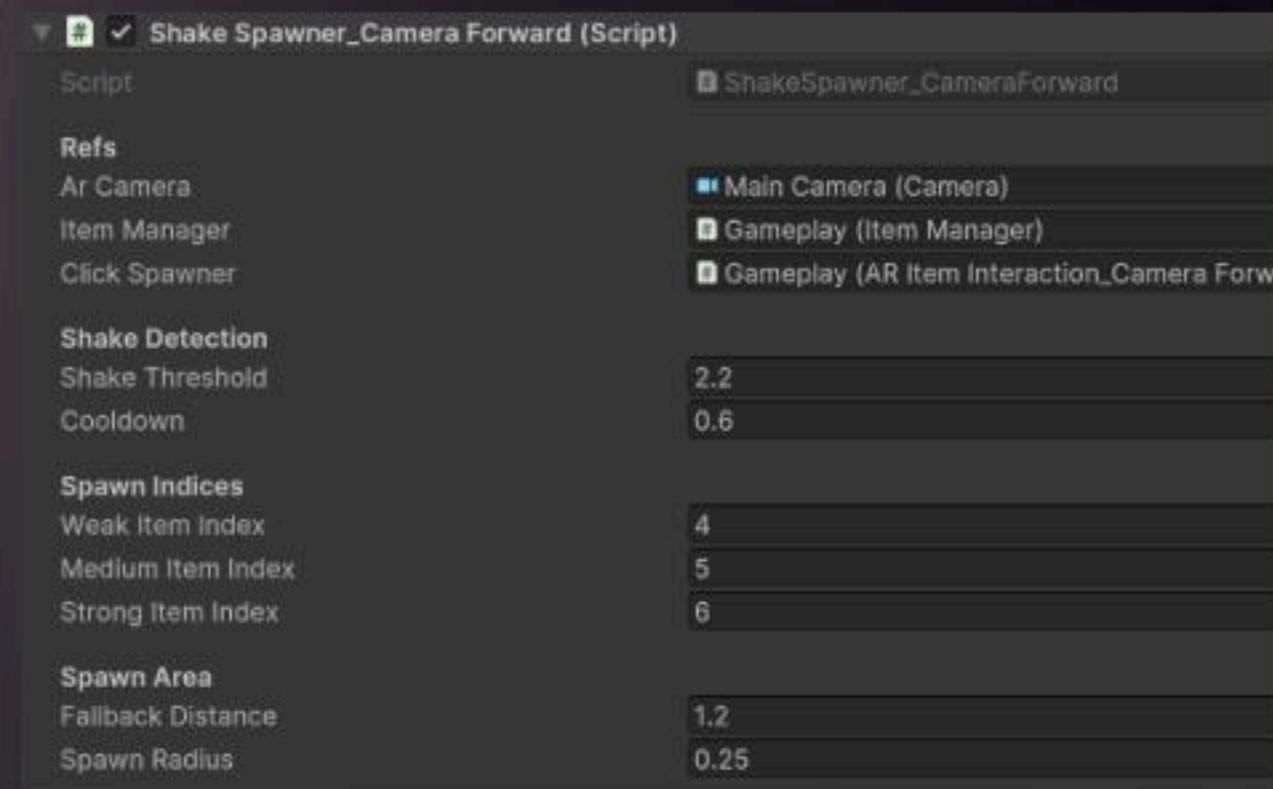
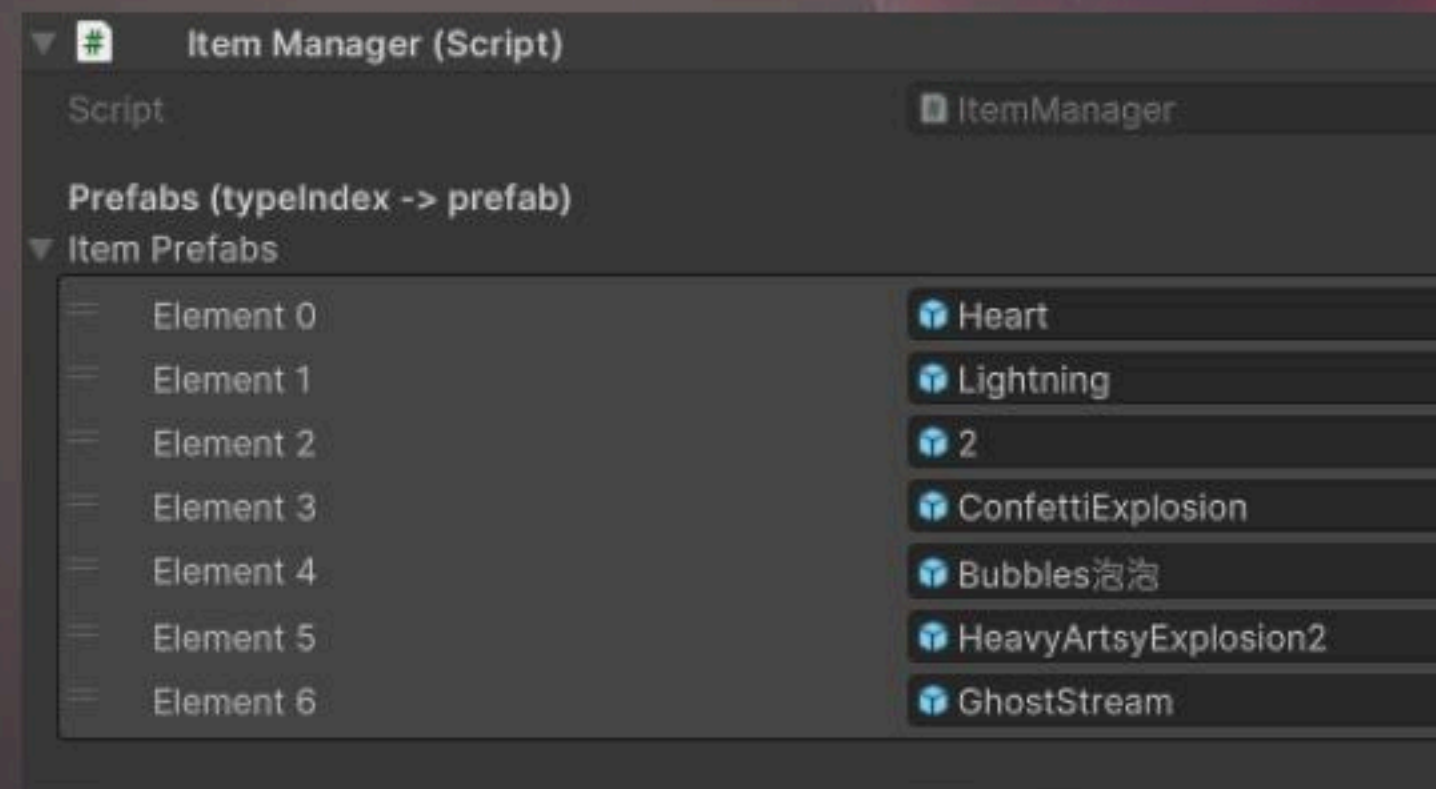
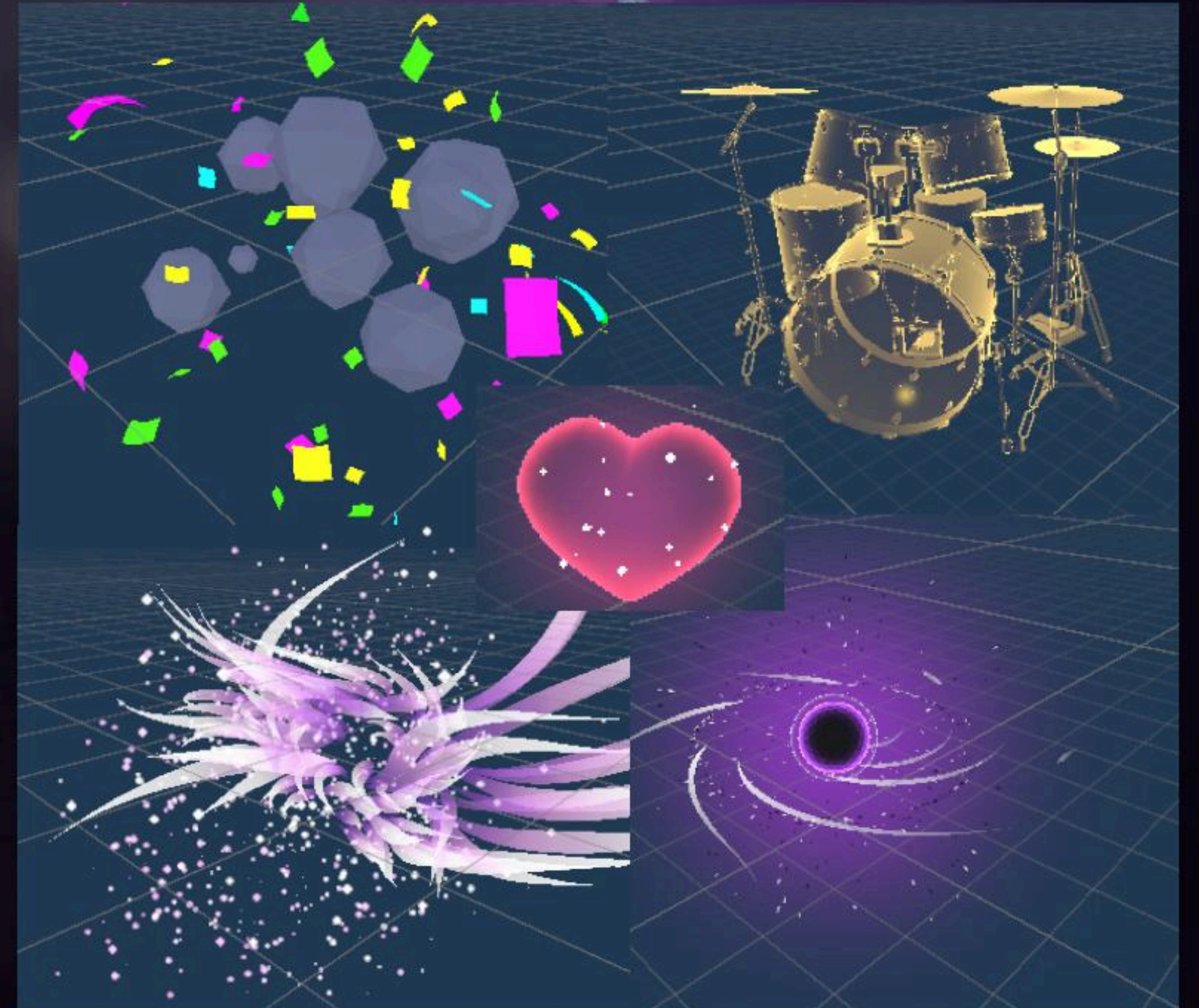


A visual change window was implemented using C# scripting, allowing you to replace new items into the project at any time.

You can also change basic attributes such as effect cooldown time and distance from the camera in the visualization window.



Prototype demonstration



Interface Design



There are two modes: 1. The app first identifies the platform (stage), and all subsequent interactions occur on the stage. 2. The app directly generates items in front of the camera.



For ease of testing and demonstration, mode 2 was ultimately chosen in the final test, but mode 1 has also been fully implemented.